Dillibabu S

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.26
Calavala Cunnan Chetty's Hindu Matriculation Higher Secondary School	Class XII (State Board of Tamil Nadu)	2019	82.83%
Calavala Cunnan Chetty's Hindu Matriculation Higher Secondary School	Class X (State Board of Tamil Nadu)	2017	93.20%

PROJECTS

Thermo-Mechanical Phase-Field Model for Elevated-Temperature Behavior of C/SiC Composites -- *M.Tech Ongoing*

- Thermodynamically consistent **phase-field fracture** formulation to incorporate manufacturing-induced residual thermal stresses and their **temperature-dependent relaxation** during reheating.
- Implemented a coupled thermo-mechanical constitutive model in **Abaqus** via a custom User Element (**UEL, Fortran**), decomposing total strain into elastic, thermal, and residual components with temperature-dependent stiffness and fracture toughness.

Damage Modelling of Machining-Induced Defects in C/SiC Composites. - *M.Tech project*

- Developed a phase-field fracture model in Abaqus (custom **UEL/UMAT subroutine, Fortran**) to simulate damage initiation and propagation, **capturing fracture pattern** under tensile, compressive, shear, and bending loads.
- Modelled hole surface defects across varying size and shape to quantify their effect on the mechanical failure behavior.
- Experimentally validated the numerical model through the stress-strain response of the C/SiC composite.

AI-Based Image Restoration for Semiconductor Inspection -- *KLA AI Hackathon*

- Built a supervised **image-restoration** pipeline to jointly denoise and super-resolve degraded grayscale semiconductor images (256px to 512px) under unknown speckle/Gaussian degradation
- Benchmarked CNN and transformer-based restoration backbones (Residual U-Net, NAFNet, SwinIR) with composite L1 + SSIM + edge-aware loss, achieving 24.63 dB PSNR /0.681 SSIM on held-out test data
- Designed a synthetic degradation-augmentation pipeline (BSRGAN-style) to improve out-of-distribution generalization, validated with domain-aware train/val splits.

Thrust Vectoring Characteristics of a Bevel Nozzle - *B.Tech project*

- Designed and built a load-cell-based force measurement rig to **experimentally quantify thrust vectoring** forces and angles.
- Simulated nozzle flow and vectoring performance in **ANSYS Fluent**, validating the numerical results against a **Schlieren imaging setup for shock flow visualization** and through image processing methods found exact angle of vectoring.
- Compared numerical vs. experimental thrust angles and forces to assess model accuracy.

Aircraft Design of a Heavy Business Jet

- Conducted conceptual design and weight estimation for a heavy business jet, validating feasibility against design requirements.
- Modelled the aircraft in CATIA V5 and estimated aerodynamic and structural loads on the wing and fuselage using ANSYS.

INTERNSHIPS

Intern, Precision Equipments (Chennai) Pvt. Ltd.

Gained hands-on experience in **heat exchanger design** and material selection; learned industry codes, QA standards,

welding techniques, and pre/post-weld heat treatment; observed Non-Destructive Testing (NDT) methods for defect detection.

SKILLS

Programming Languages: Fortran, Python (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn), MATLAB

Technical Skills: Data Structures & Algorithms, Machine Learning

Design & Simulation Software: CATIA V5, Solid Edge, ABAQUS (subroutine development), ANSYS (Static Structural, Fluent), NASTRAN (Linear, Nonlinear), Hypermesh

SCHOLASTIC ACHIEVEMENTS

Presented **"Thermo-mechanically Coupled Phase-Field Modelling of Defect-Driven Tensile Failure in C/SiC Composites at Elevated Temperatures."**

-IMECE India 2026 (ASME), Chennai Trade Center, Chennai.

Developed a thermo-mechanical phase-field model to predict how matrix cracks, surface flaws, and geometric defects drive tensile failure in C/SiC composites at high temperature, validated against published experimental data.

Presented **"Effect of Machining-induced Defect on the Failure Behaviour of C/SiC Composite"**

-IMECE India 2025 (ASME), HICC Hyderabad.

Used phase-field modelling to study drilling-induced defects and their effect on failure behavior of aerospace-grade composites.

CERTIFICATIONS

NPTEL: Introduction to Composites, Introduction to Aircraft Design, Digital Image Processing.

NPTEL: Fundamentals of combustion, Fluid Dynamics in Astrophysics.

Short-term course: "Application of Ionizing & Non-Ionizing Radiation for NDE" -- IGCAR Kalpakkam

Composites Fabrication Repair - MIT, Chrompet

NxtWave Launchpad: DSA Summer Sprint (Cohort Completion)

RELEVANT COURSES

Mathematical Methods for Engineers, Advanced Mechanics of Solids, Advanced Finite Element Methods, Fracture Mechanics, Analysis and Design of Composite Structures, Linear Algebra and Vector Calculus.

Machine Learning and its Application

Structures, Strength of Material, Heat Transfer, Aerodynamics, Propulsion, Flight Dynamics.

POSITION OF RESPONSIBILITY

PG Buddy, Sunshine -- The Counselling Cell of IIT Hyderabad; provided peer mental health support and coordinated campus wellness events.

EXTRACURRICULAR

Represented IIT Hyderabad in Athletics (Javelin Throw) at the 58th Inter-IIT Sports Meet.

Won 1st place in the Yoga Competition held at IIT Hyderabad on the occasion of International Yoga Day 2026.

HOBBIES AND INTEREST

Playing bugle and flute, learning new language (currently Japanese and Telugu), Watching global news